

Wrocław University of Science and Technology
Faculty of Management

Business Intelligence Workplace

First task set report

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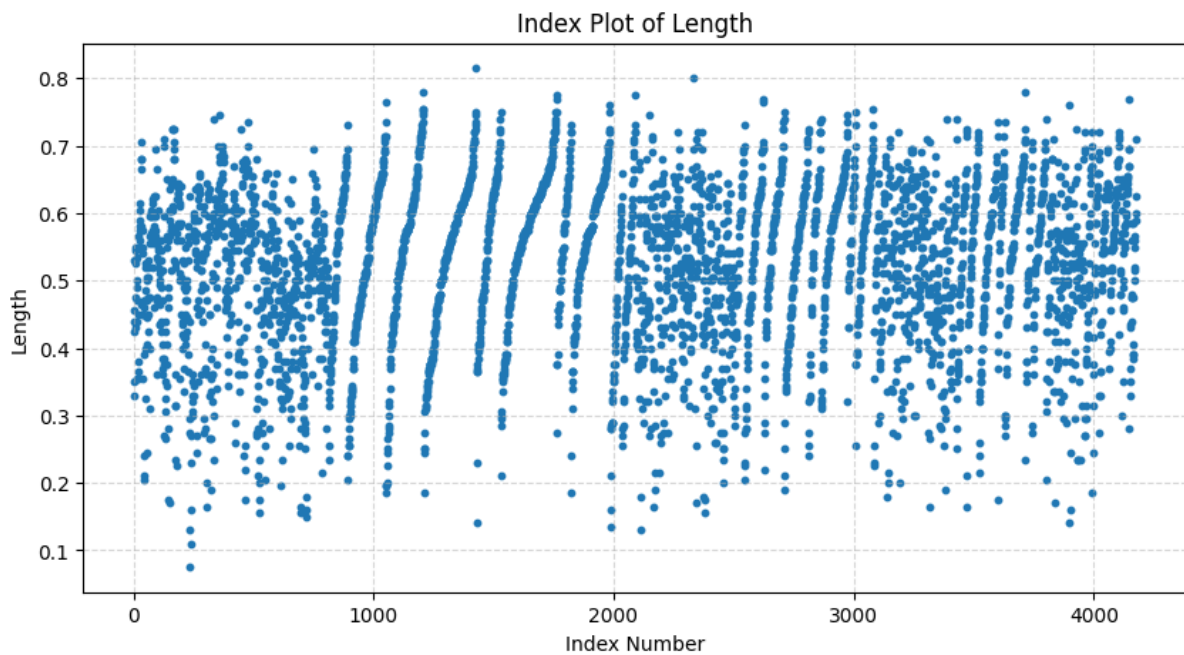
dr Arkadiusz Jędrzejewski

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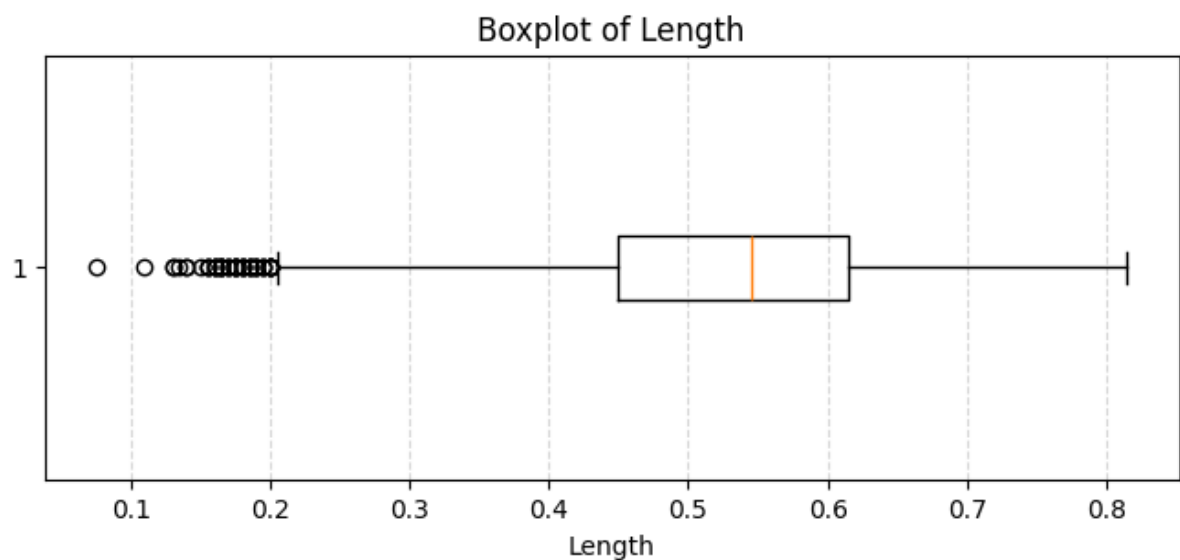
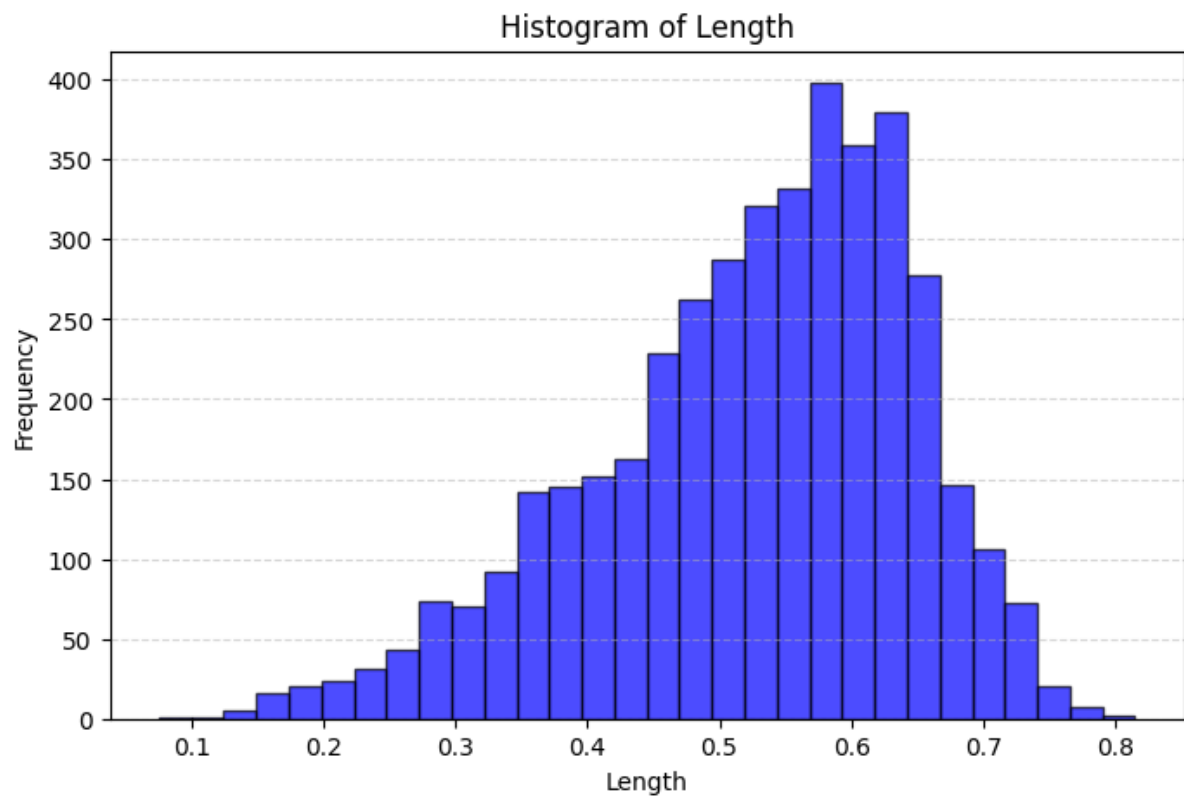
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1 Index plots for length and whole weight

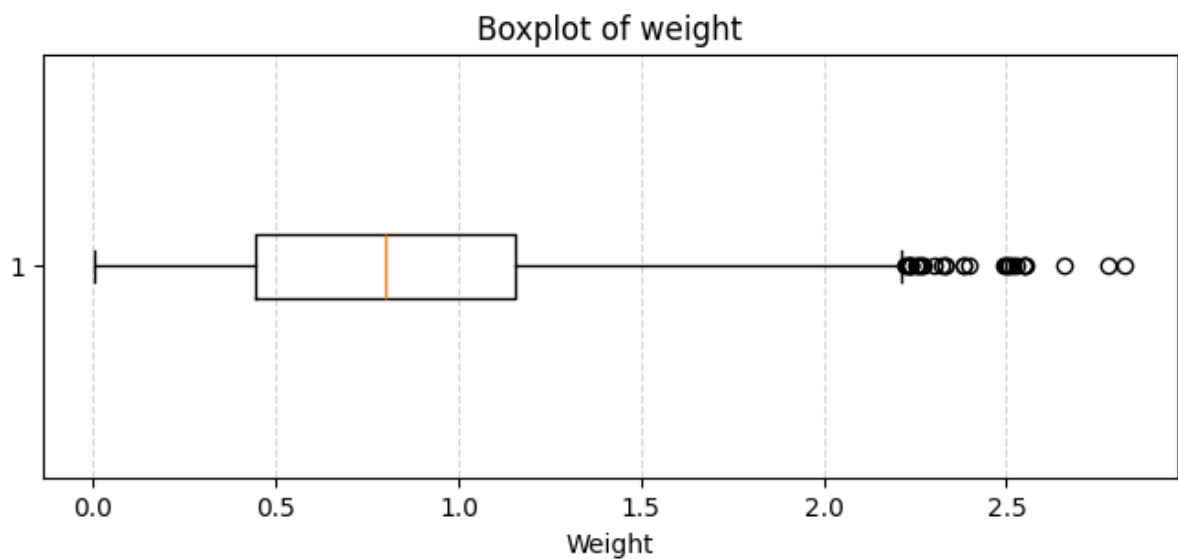
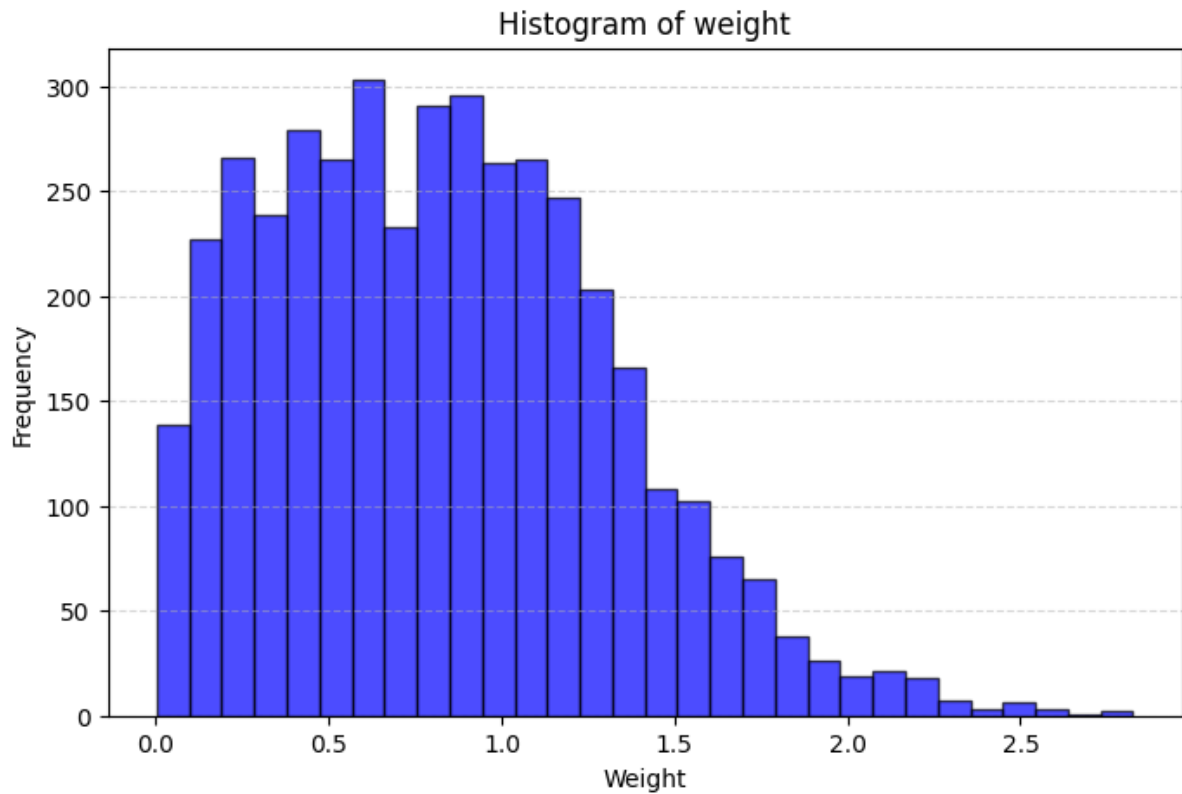


Both of index plots presented have dense scatter points throughout their grids. With that said, we can initially reject a hypothesis, that there is some seasonality in the data. Such dense scatter points suggest that the data rows are rather reasonably mixed and diversified. The only fragments where a slight trend is visible, are the horizontal, slightly skewed lines which probably were born as a result of uploading the dataset in large batches.

2 Boxplots and histograms for both the variables



Both of these plots visualizing characteristics of the "Length" attribute are presenting generally normally distributed data, with center between values 0.5 and 0.6. Moreover, this distribution is slightly left-skewed, as the boxplot shows. Most of the data is located to the left of the distribution center, as well as the dataset's outliers.



Plots visualizing characteristics of the "Weight" attribute are a complete opposite of previously described "Length", being strongly right-skewed, with many outliers to the right of the normal distribution's center.

3 Descriptive statistics

min	0.075	# length descriptive statistics
Q1	0.450	
Median	0.545	
Q3	0.615	
Max	0.815	
IQR	0.165	

The "Length's" attribute descriptive statistics are proving previous theses right, as well as the following "Weight's" attribute descriptive statistics:

min	0.0020	
Q1	0.4415	
Median	0.7995	<i># weight descriptive statistics</i>
Q3	1.1530	
Max	2.8255	
IQR	0.7115	

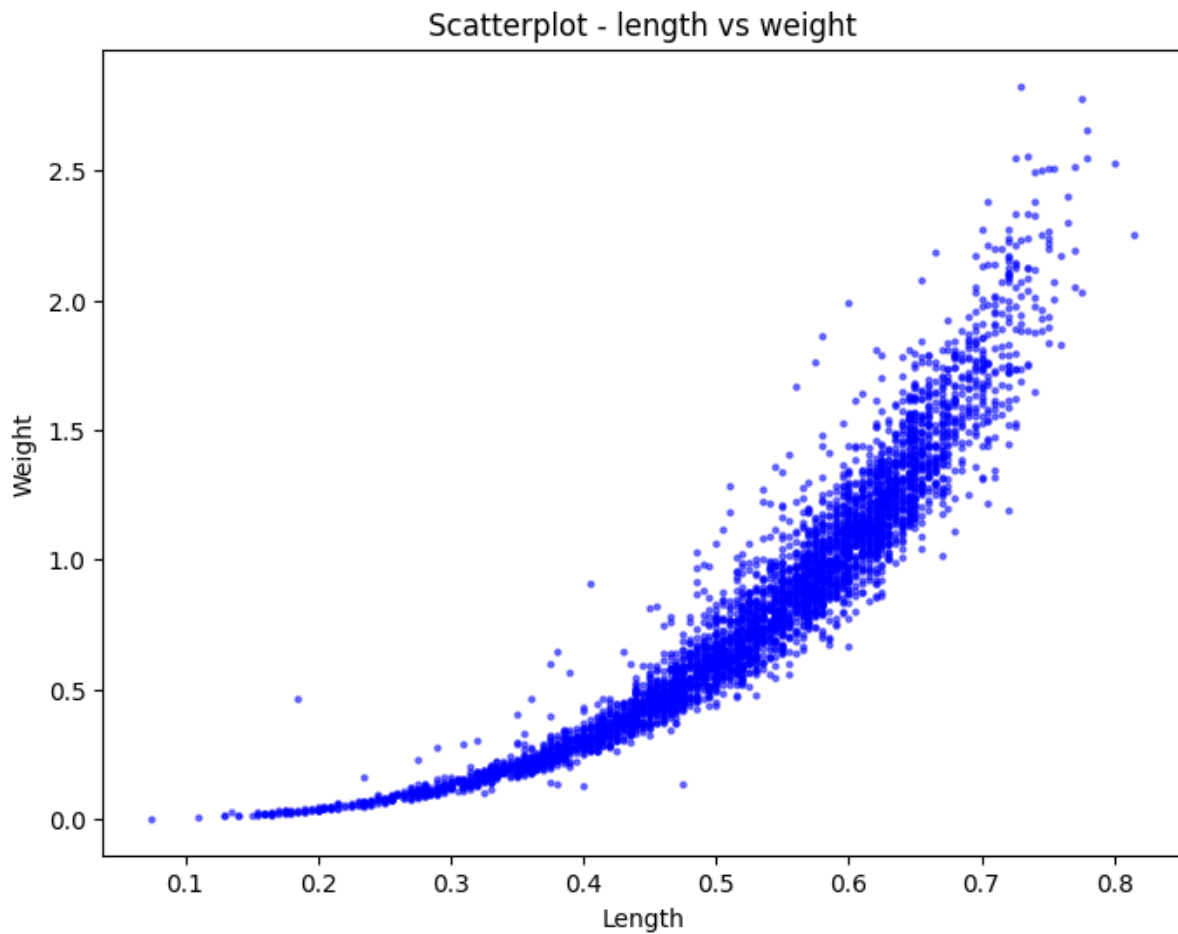
4 Moment-based statistics

mean	0.523992	
sd	0.120093	
variance	0.014422	<i># length moment-based statistics</i>
kurtosis	0.064621	
skewness	-0.639873	

mean	0.828742	
sd	0.490389	
variance	0.240481	<i># weight moment-based statistics</i>
kurtosis	-0.023644	
skewness	0.530959	

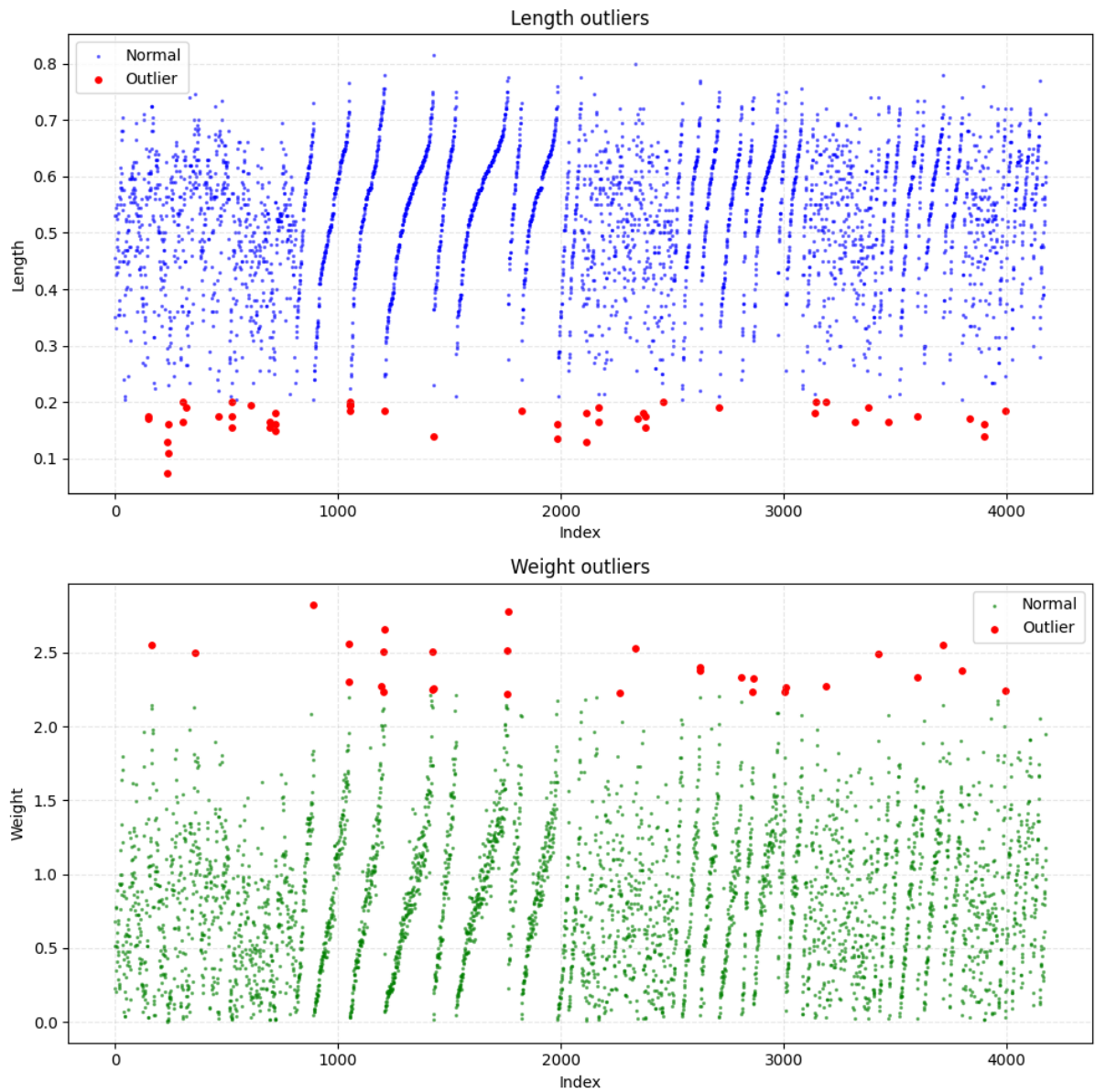
Both of these outputs further confirm previously established theses about datasets being skewed. The only notable difference between these attributes' characteristics are the values of variance and standard deviation - in case of "Weight" it is notably bigger, which means that its values are indeed scattered.

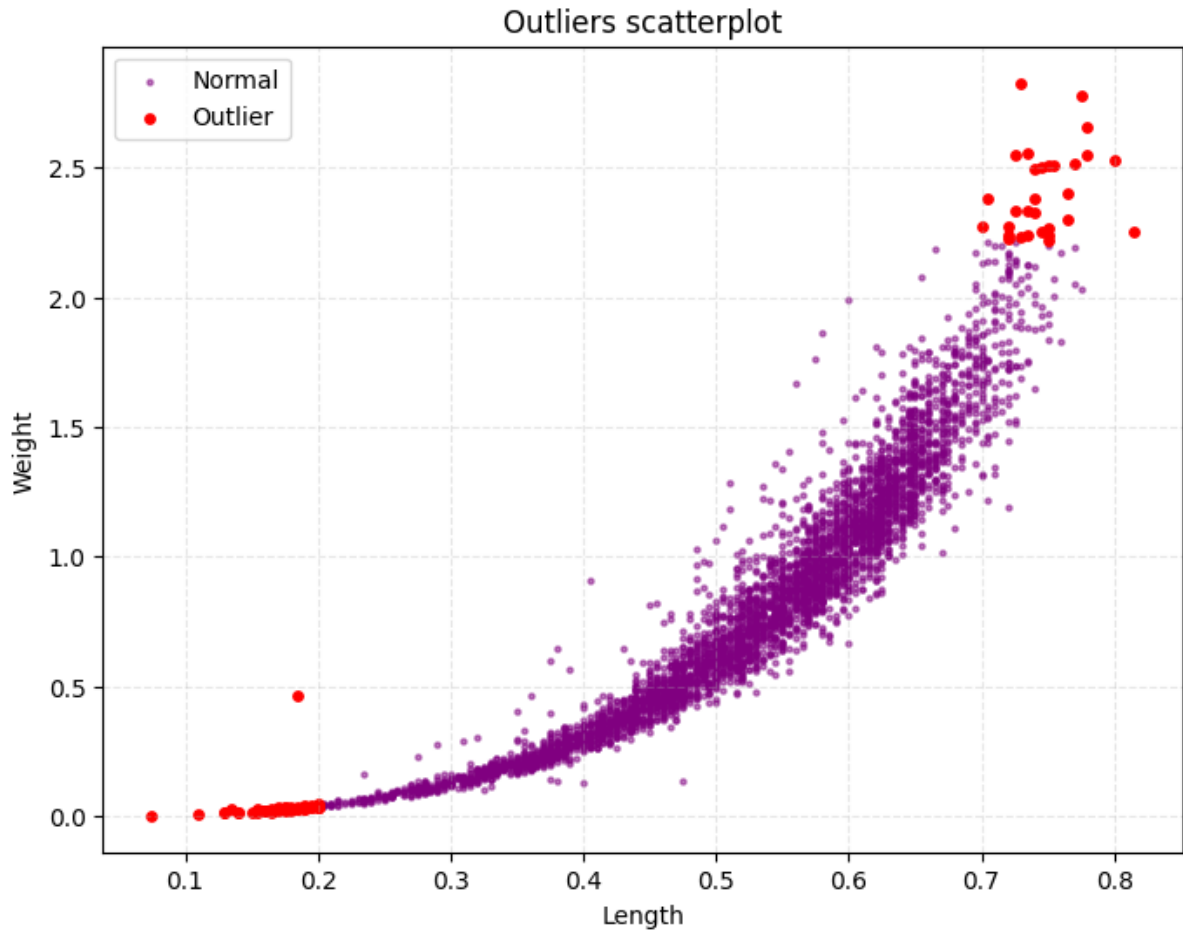
5 Scatterplot of whole weight vs length



The scatterplot reveals a strong and positive relationship between the "Length" and "Weight" attributes. As the "Length" increases, the "Weight" increases as well at an accelerating rate. This exponential curve is expected, as weight is proportional to volume, which scales roughly with the cube of the length.

6 Outlier identification and removal

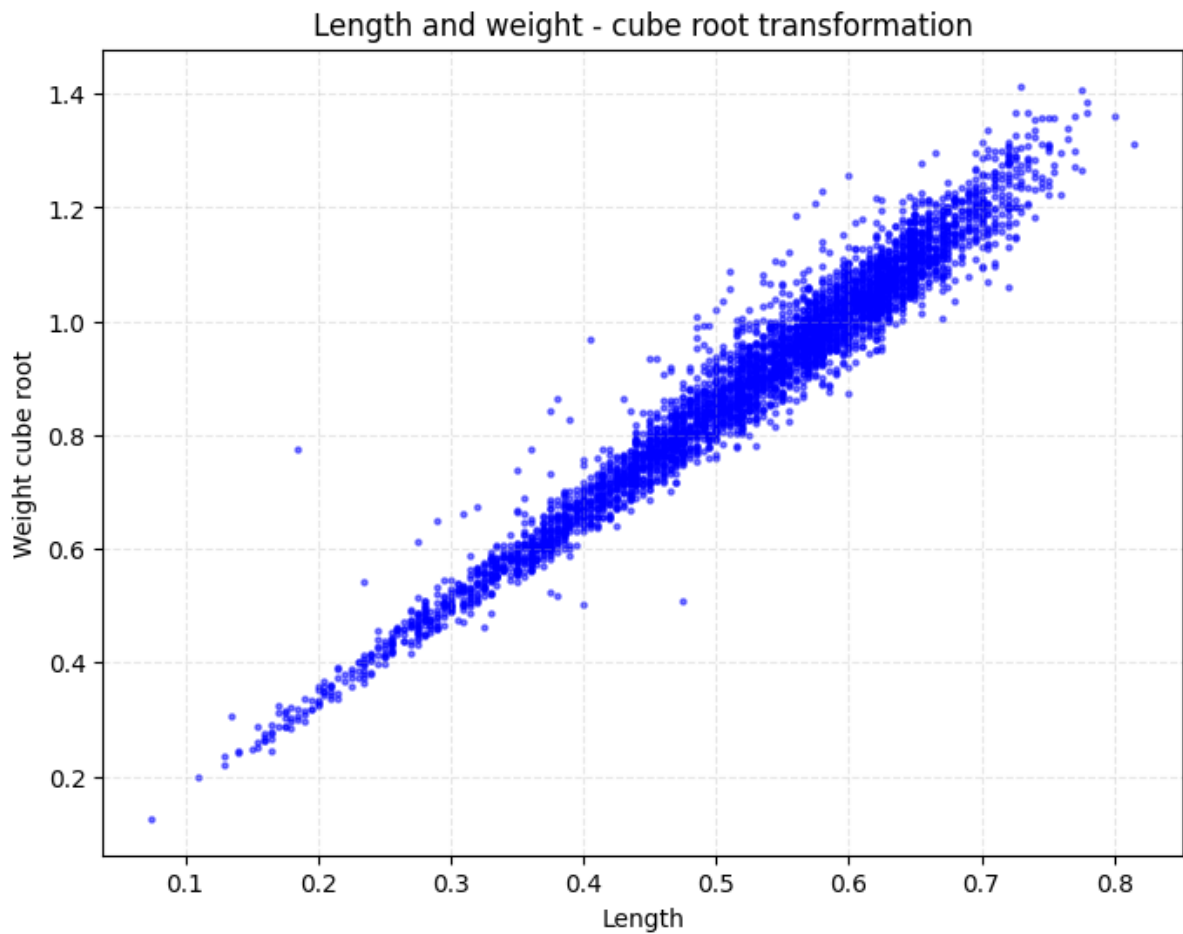




Outliers were identified using the IQR method, classifying any value falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ as an outlier. A combined total of 79 rows were identified as outliers in both of the variables and removed from the dataset in the same time. The descriptive statistics for the cleaned dataset are presented below:

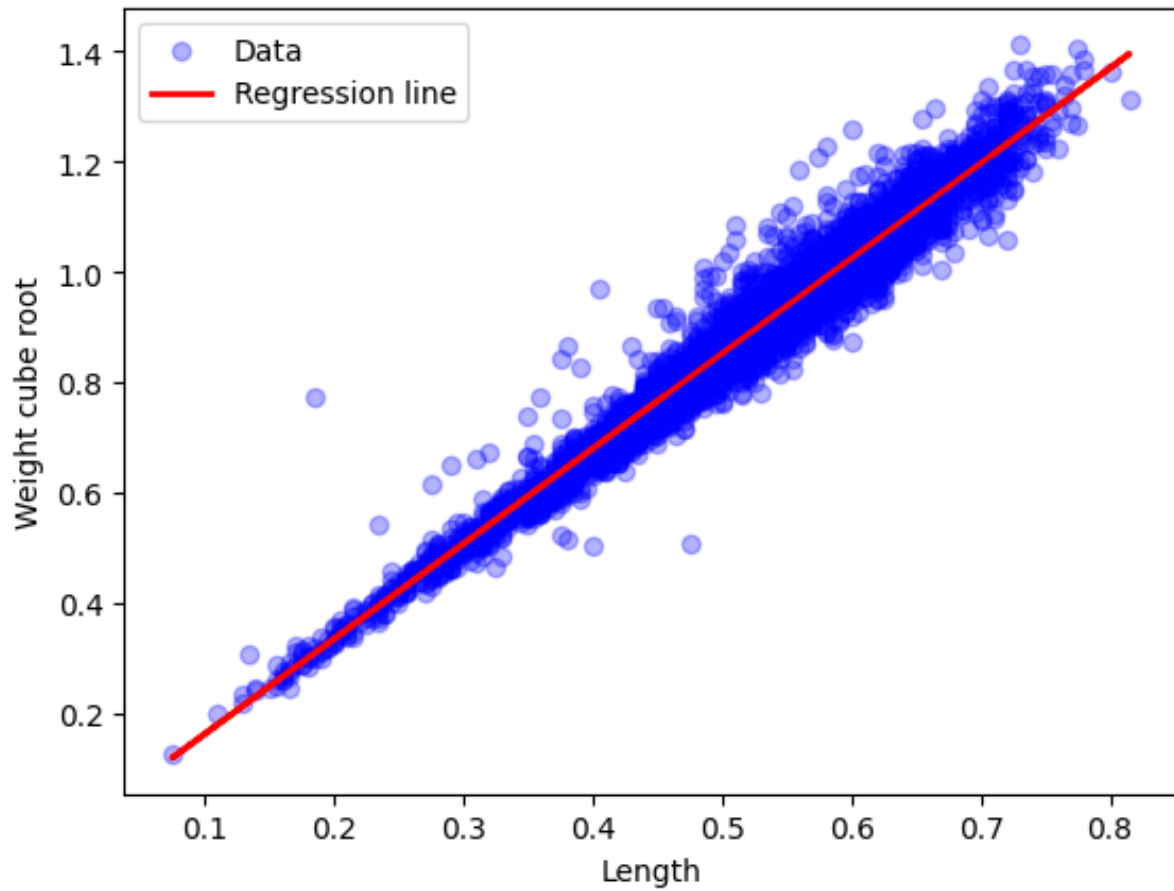
	Min	Q1	Median	Q3	Max	IQR
Length	0.2050	0.4550	0.5450	0.615000	0.775	0.160000
Weight	0.0385	0.4495	0.8025	1.149125	2.210	0.699625

7 Data transformation



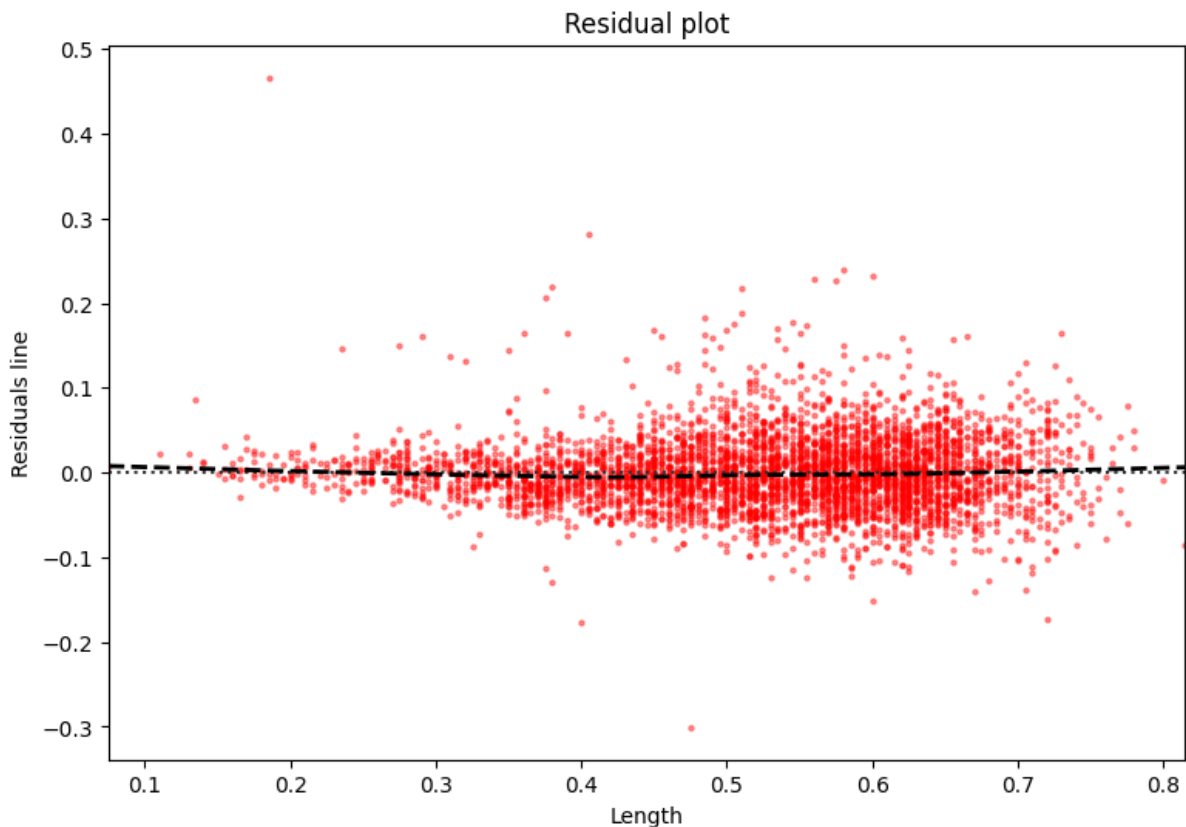
To apply linear regression, the non-linear exponential trend observed in the previous scatterplot required transformation into a straight lined trend. Two transformations were computed - plotting "Length" against the Cube Root of "Weight" and applying a natural logarithm to both variables. In the end the Cube Root transformation was selected for the final regression model.

8 Regression modeling



A linear regression model was fitted to the "Length" attribute and the Cube Root of the "Weight". The resulting model yielded an exceptionally high coefficient of determination being of 0.9585 value, indicating that approximately 95.85% of the variance in the transformed weight is explained by the "Length". The MSE being extremely low signifies that there is a tight fit.

9 Error visualization



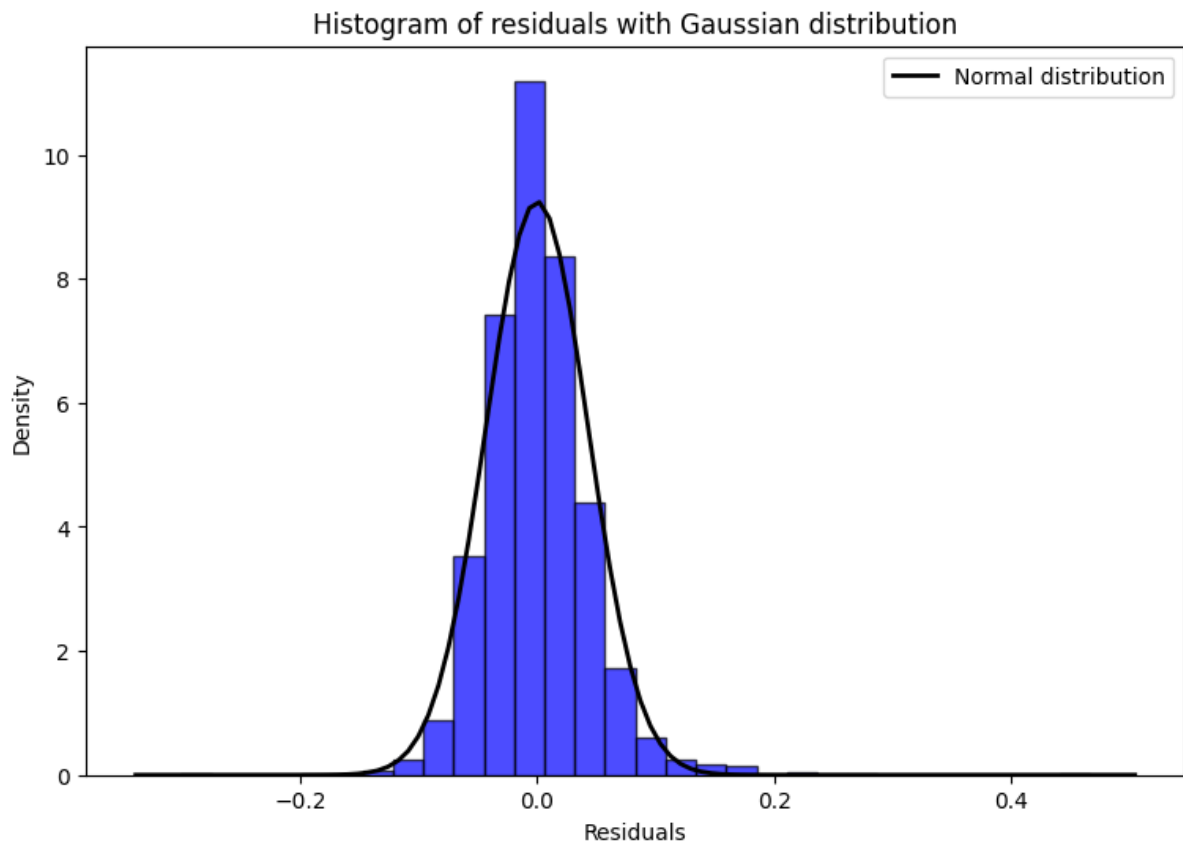
The residual plot indicates that the variance of the errors is not constant (because as the length increases data points start being all over the place). As the predicted values increase along the x axis, the spread of the residuals widens significantly, suggesting the model is less precise when predicting the larger weights. However, the calculated mean of the residuals was exactly zero, with a standard deviation of 0.0431 and a variance of 0.00186.

10 Statistical error check

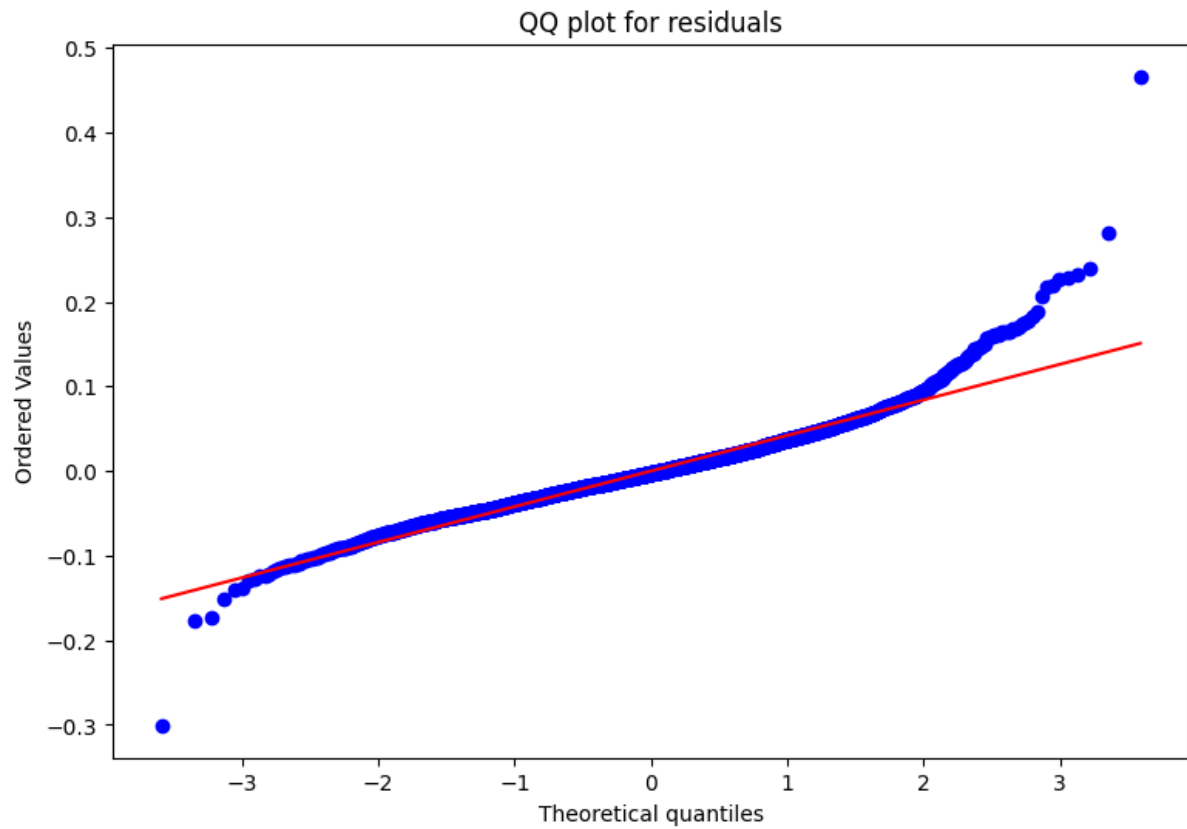
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Mean of the residuals = 1.0472297622751057e-16
Standard deviation of the residuals = 0.04315328939871089
Variance of the residuals = 0.0018626523165768648
```

The mean of the residuals evaluates to 1.047×10^{-16} , which is effectively zero - this is a fundamental mathematical property of OLS linear regression, confirming that the regression line is perfectly centered through the data and that the model's predictions are - on average - completely unbiased. The standard deviation of roughly 0.0432 indicates that the model's predictions for the cube root of the whole weight typically deviate from the true values by only a little.

11 Normality analysis

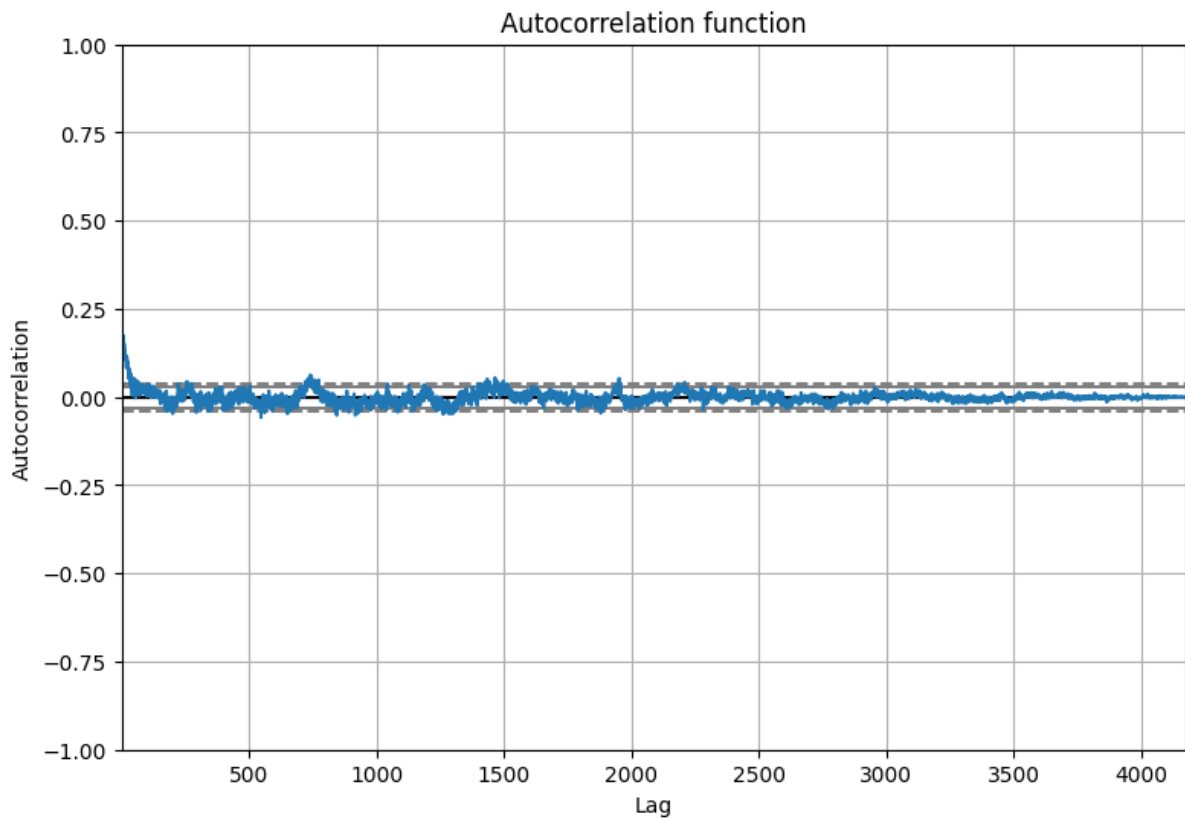


As mentioned earlier, while the histogram of the residuals generally follows the bell shape of a normal distribution curve, it is notably highly peaked around the center. This indicates that the data is more concentrated around the mean than a standard normal distribution would predict.



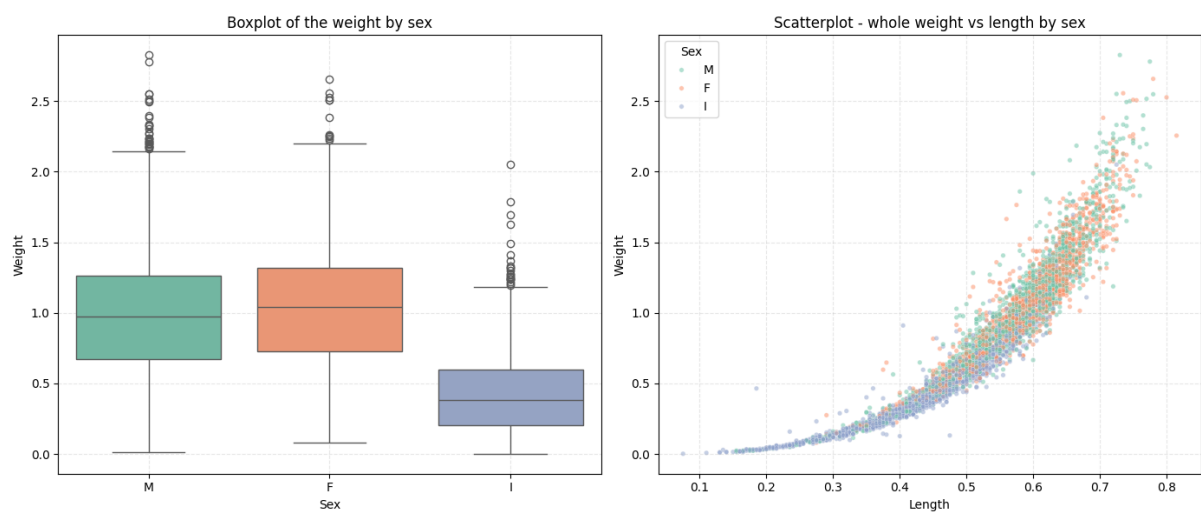
The QQ plot further confirms the departure from perfect normality. The data points deviate significantly from the theoretical straight red line at both the lower and upper extremes. These deviations indicate that the residuals contain more extreme outlier values than would be expected in a perfectly normal distribution (as mentioned in an initial hypothesis).

12 Independence test



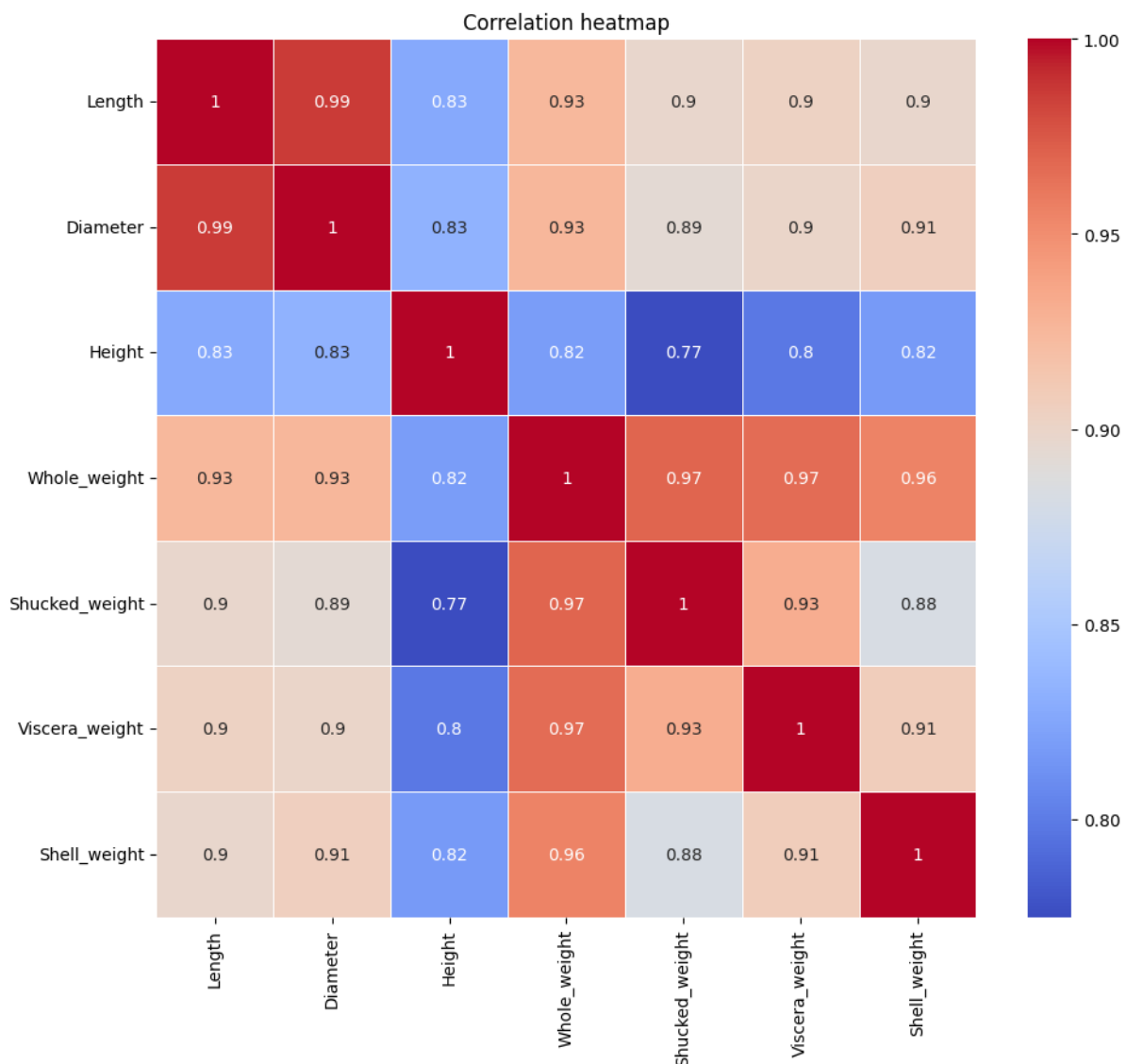
To test for the independence of the residuals, an autocorrelation plot was generated. Aside from a very minor initial spike at the earliest lags, the autocorrelation line remains tightly bounded within the horizontal confidence intervals across the entire grid. This confirms that the residuals are independent and do not exhibit any meaningful autocorrelation, which is expected given that the data points represent individual, independent measurements rather than a sequential time series.

13 Group segmentation



Segmenting the data by the categorical "Sex" attribute reveals major differences between infants and adults. The boxplot demonstrates that infants have a significantly lower median weight compared to adults. The segmented scatterplot further illustrates this growth trajectory. Infants predictably dominate the lower left quadrant of the plot, while adult males and females are heavily intermixed in the upper-right quadrant. This indicates that while physical dimensions can easily differentiate infants from adults, they are practically insufficient for distinguishing between adult males and females.

14 Correlation analysis



In the correlation heatmap, we can see strong positive relationships across all physical attributes. Notably, the Length and Diameter attributes exhibit an almost perfect correlation coefficient of 0.99. Similarly, "Weight" correlates very strongly with its components, particularly "Shucked_weight" (0.97) and "Viscera_weight" (0.97). Among the primary physical dimensions, "Height" shows the weakest relative correlation with the weight components, dropping to 0.77 when compared against "Shucked_weight".

15 Final remarks

The comprehensive exploratory data analysis and statistical modeling conducted in this project reveal several clear patterns regarding the physical attributes of abalones within the dataset. Conducted tests and visualizations have shown that there is a highly predictable and positive relationship between the physical dimensions of the subjects. Because "Weight" scales exponentially with "Length", applying a cube root transformation to the "Weight" allowed for a highly effective linear regression model, which explains approximately 95.85% of the variance in the data. However, the residual analysis highlights a few limitations - while the errors are independent and perfectly centered around zero, they tend to be not constant across all levels. The model's predictive precision decreases as the size of the abalone increases, and the distribution of errors contains more extreme outliers than a standard normal distribution would assume. Also, the correlation analysis confirms really good multicollinearity among the primary physical dimensions, particularly between "Length" and "Diameter" with 0.99 correlation value. Furthermore, while these physical attributes are excellent indicators for distinguishing infants from adults, they are practically insufficient for distinguishing between adult males and females, as their growth trajectories heavily overlap. Finally, the cube root transformation proved to be a highly effective method for linearizing the data, and the resulting model serves as a strong baseline.